

CHAPTER 7

MOMENTUM

1.6 Momentum

- 1 Define momentum as mass $\times$ velocity; recall and use the equation
 $p = mv$
- 2 Define impulse as force $\times$ time for which force acts; recall and use the equation
impulse $= F\Delta t = \Delta(mv)$
- 3 Apply the principle of the conservation of momentum to solve simple problems in one dimension
- 4 Define resultant force as the change in momentum per unit time; recall and use the equation

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time taken}}$$

$$F = \frac{\Delta p}{\Delta t}$$

1. O/N/23/12/Q10

Which expression is equal to impulse?

- A** change in momentum $\div$ time
- B** change in momentum
- C** force $\times$ distance moved in the direction of the force
- D** force $\div$ area

2. M/J/23/11/Q10

A constant frictional force acts on a moving object for a short amount of time causing the momentum of the object to decrease.

Which expression is used to calculate the time for which the force acts?

- A** time = $\frac{\text{change in momentum}}{\text{force}}$
- B** time = change in momentum $\times$ force
- C** time = $\frac{\text{momentum}}{\text{force}}$
- D** time = momentum $\times$ force